Spectrally Parameterized Neural Inverse Reconstruction

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Abstract

In neural inverse reconstruction, the forward model is typically treated as a fixed simulator that maps a neural scene representation to measurements. This work studies how the parameterization of the forward operator itself shapes the optimization landscape of coherent inverse problems. We introduce *Moray*, a reconstruction framework built on spectrally parameterized forward operators that bypass conventional time-domain synthesis and instead evaluate measurements directly through closed-form spectral kernels. This parameterization induces a substantially shorter and better-conditioned differentiable graph compared to the time-domain approaches. We further introduce *Phase-Coherent Manifold Parameterization*, a scene representation that jointly learns scene reflectivity and a deformable surface manifold, allowing the reconstruction to adapt to geometric deviations from the assumed imaging plane and thereby reducing common artifacts in coherent reconstruction. We instantiate these ideas for 77 GHz radar imaging using real measurements from a synthetic-aperture mmWave system. Across multiple challenging scenes, Moray consistently improves reconstruction fidelity, suppresses background artifacts, and degrades more gracefully under data limitations. More broadly, the results suggest that, in neural inverse problems, exposing appropriate analytical structure of the sensing physics to the optimizer can be as important as the choice of neural representation itself.

1 Introduction

Neural Inverse Reconstruction and Forward Models. Many recent reconstruction methods formulate imaging as an analysis-by-synthesis problem: a neural scene representation is optimized so that measurements synthesized through a differentiable forward model match the observations. This paradigm underlies neural rendering and surface reconstruction methods and has increasingly been extended to physical sensing modalities including transient imaging, ultrasound, RF propagation, and coherent acoustics. In these systems, the forward model encodes the sensing physics that maps the scene field to measurements through propagation, scattering, sampling, and signal processing operations. As a result, optimization behavior depends not only on the neural representation itself, but also on how this physical operator is represented inside the differentiable pipeline. This becomes particularly important in coherent sensing systems such as radar, sonar, and optical coherence tomography, where measurements retain carrier-phase information and the forward model operates on highly oscillatory complex-valued signals.

Limits of Conventional Parameterizations. Most existing differentiable sensing pipelines implement the forward model by directly mirroring the physical acquisition process. In coherent sensing systems, this typically involves synthesizing a time-domain signal by integrating scene responses against the transmitted excitation, sampling the resulting waveform, and finally applying spectral transforms such as the discrete Fourier transform before evaluating the reconstruction loss. While this

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